

Step 172 Track Closeout: RH Membrane Packet Cascade

Manager-led arc steps 163–172

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1 Scope

This packet is written at

/home/repos/six-birds-foundations-iii/anti_loc/thread/step172_track_closeout_artifacts/.

It is a final track closeout admitted under the synthesis-stacking memory case for final closeout. It composes earlier verdicts and does not supply any external analytic record.

2 Cascade Typed State

The primary parent residual is

$$\Xi_{BC} = B_{l,a}^* \Pi_{Y_a} B_{l,a}$$

on the inherited Burnol/Sonine pulled-evaluator carrier. The sibling parent residual is Ξ_{BC}^{Hecke} on the Hecke L -function carrier through step 92.

Branch A: Calkin bridge

Branch A has status `split_external_theorem`. Its typed context is

$$A_\eta = C^*(P_\infty, M_{m_l}, P_\eta, I), \quad K_\eta = A_\eta \cap K(H), \quad q_\eta(C_l P_\eta) \in A_\eta / K_\eta.$$

The open gates are G2 faithful boundary symbol σ_B , G3 normal form $C_l = U_l W_l + K_l$, G4 lower-faithfulness of U_l , and G5 compact remainder record for K_l .

Branch B: Per-zero finite carrier

Branch B has status `finite_carrier_diagnostic_indeterminate`. Its typed context is

$$H_\eta = \widehat{\bigoplus_\rho} H_{\eta,\rho}, \quad A_{\eta,\rho} \subset A_\eta.$$

The active sub-residual is

$$\Xi_{\text{matrix_source}_{l,\text{Rho_fin},A_fin,K_fin}}.$$

The open external content is SL164.1, exact projected Burnol/Sonine kernel evaluator entries, currently only support-level.

Branch C: Shifted co-Poisson

Branch C began as the H1–H5 shifted co-Poisson branch and was upgraded in steps 169–171. Step 169 named

$$\text{ZI-COV}_{l,a}^{CP} : \begin{cases} (i) & P_\infty M_\zeta G = M_\zeta A_\infty G, \\ (ii) & T_a P_\infty M_\zeta H = M_\zeta A_{T,a} H. \end{cases}$$

Step 170 proved the zero-free output-support subclass for (i):

$$A_\infty^\Omega G := M_{1/\zeta} P_\infty M_\zeta G$$

on

$$\mathcal{G}_\Omega = \{G \text{ legal} : \text{supp}(P_\infty M_\zeta G) \subset \Omega, \inf_{s \in \Omega} |\zeta(s)| > 0\}.$$

The target-equivalence test was not target-equivalent under inherited records: P_∞ is the archimedean Sonin/prolate projection, not a zeta-zero spectral projection. Step 171 left projected zero-jet surjectivity unresolved in the route table and kernel-required in the schema: the missing record is a Mellin-side Sonin/prolate kernel $K_\infty(s, s')$, or an equivalent theorem deciding

$$\langle M_\zeta G, P_\infty y_{\rho,k} \rangle$$

for legal Burnol G .

Hecke sibling route

Step 168 audited the H6 descent from the Hecke fiber $K = \mathbb{Q}, \chi = 1$ to the Burnol/Sonine carrier and returned `non_comparability` under inherited records. The scalar equality $L_{\mathbb{Q}}(s, 1) = \zeta(s)$ is not a carrier identity.

3 Closeout Theorem

Theorem 1 (RH membrane packet cascade closeout, manager-led arc steps 163–172). *On the inherited Burnol/Sonine pulled-evaluator carrier through step 162 and the Hecke L -function carrier through step 92, the parent residual Ξ_{BC} and sibling residual Ξ_{BC}^{Hecke} have the following typed cascade structure.*

- (i) *The three open child branches of Ξ_{BC} are typed at the analytical-structural level: Branch A is the Calkin-bridge G2–G5 route, Branch B is the per-zero SL164.1 finite-carrier route, and Branch C is the shifted co-Poisson H1–H5 route upgraded by subclass and keystone audits.*
- (ii) *Branch C admits a conditional subclass factorization. On the joint zero-free subclass $\mathcal{G}_{\Omega, \Omega'}$, where $P_\infty M_\zeta G$ avoids zeros in Ω and the corresponding $T_a P_\infty M_\zeta$ output avoids zeros in Ω' , the operator $A_\infty^\Omega = M_{1/\zeta} P_\infty M_\zeta$, together with an externally supplied $A_{T,a}^{\Omega'}$ for ZI-COV(ii), gives the template*

$$M(B_{l,a} C g)(s) = \zeta(s) A_{T,a}^{\Omega'} \left(m_l(G - A_\infty^\Omega G) \right) (s)$$

on that joint subclass.

- (iii) *The keystone target-equivalence question, projected zero-jet surjectivity of P_∞ , is not decided under inherited records. The specific missing record is $K_\infty(s, s')$, or an equivalent range-evaluator matrix theorem deciding $\langle M_\zeta G, P_\infty y_{\rho,k} \rangle$.*

- (iv) *The Hecke route to Ξ_{BC} closure through zeta-fiber descent at $K = \mathbb{Q}, \chi = 1$ is foreclosed under inherited records by step 168 non-comparability.*
- (v) *The retained no-gos remain active: public-shadow non-promotion, finite-window Calkin blindness, auxiliary-GRH smuggling, incomplete character spectrum support-only, scalar L -function identity is not carrier identity, and ledger-relative Hecke non-comparability.*

Consequently, the RH membrane packet at this carrier is

cascade_diagnostic_complete_at_inherited_records_ceiling.

Proof. Item (i) is the composition of steps 163–166. Branch A is inherited from the scoped Calkin bridge, Branch B from the per-zero finite-carrier diagnostic, and Branch C from the shifted co-Poisson obligation split. Item (ii) composes the step 169 reduction to ZI-COV with the step 170 zero-free output-support construction for ZI-COV(i), while leaving ZI-COV(ii) as an external entry point. Item (iii) is the step 171 audit: every constructive or refutation path reduced to the same missing K_∞ or range-evaluator theorem. Item (iv) is exactly the step 168 bridge audit. Item (v) lists the retained framework no-gos carried forward through the arc. No gate is proved in this closeout; the conclusion is only the typed cascade verdict at the inherited-record ceiling. $\square$

4 External-Content Interface

Branch A.

Typed context: A_η/K_η with $q_\eta(C_l P_\eta)$. Required record: G2–G5 Calkin-faithful boundary-symbol theorem. Expected verdict shape: a carrier-grade Calkin verdict route for $q_\eta(C_l P_\eta)$ and the corresponding Ξ_{BC} branch decision. No-gos respected: public-shadow non-promotion and finite-window Calkin blindness.

Branch B.

Typed context: $H_\eta = \hat{\oplus}_\rho H_{\eta,\rho}$ and $A_{\eta,\rho} \subset A_\eta$ on the declared finite carrier. Required record: SL164.1 exact projected Burnol/Sonine kernel evaluator entries. Expected verdict shape: decision of $\Xi_{\text{matrix_source}}$ and activation map for exact coupling or summability records. No-gos respected: finite-window Calkin blindness and public-shadow non-promotion.

Branch C-K.

Typed context: Burnol/Sonine Mellin carrier with P_∞ the transported Sonin/prolate projection and zero-evaluator jets $y_{\rho,k}$. Required record: $K_\infty(s, s')$ or equivalent range-evaluator theorem. Expected verdict shape: decision of projected zero-jet surjectivity and target-equivalence status for Branch C.

Branch C-II.

Typed context: $T_a P_\infty M_\zeta$ output subclass avoiding zeros in Ω' . Required record: $A_{T,a}^{\Omega'}$ satisfying ZI-COV(ii) on that subclass. Expected verdict shape: joint zero-free subclass factorization using A_∞^Ω and $A_{T,a}^{\Omega'}$.

5 Framework-Derived Outputs

Predictive structural outputs are the retained no-gos. Analytical structural theorem-grade outputs are the step 163 scoped residual theorem, step 166 cascade summary, step 167 Hecke pivot

declaration, step 168 Hecke non-comparability typed no-go, steps 169–170 conditional subclass factorization on Branch C, and step 171 keystone audit. Finite-carrier diagnostics are step 164 and the sidelined CP-RED bump-generator experiment at

/home/repos/six-birds-foundations-iii/anti_loc/thread/.step170_cp_red_test_sidelined/.

Organizational/audit outputs are the residual trees, bridge atlases, route status tables, nonclaim boundaries, and validation scripts.

6 Retrospective and Handoff

The manager-led arc produced a typed cascade with named external interfaces, not a closed proof. Its substantive forward pushes were the Hecke non-comparability audit and the Branch C reductions through ZI-COV, zero-free subclass division, and projected zero-jet audit. Subsequent work must supply one named external record or pivot to a different primary carrier outside the Burnol/Sonine and Hecke cascades of this arc.